Social Media Engagement Analytics

Using Python

Objective

To clean, transform, analyze, and visualize social media engagement data (likes, comments, shares, impressions, watch time, followers) using Python, and generate actionable insights for content optimization.

Analysis Process

1. **Data Import & Cleaning** → Handled missing values, duplicates, standardized categories, corrected unrealistic values.
2. **Feature Engineering** → Created engagement score, engagement rate, hashtag count.
3. **Exploratory Analysis** → Summary statistics, correlation matrix, groupby summaries.
4. **Statistical Profiling** → Mean, median, mode, variance, skewness, kurtosis.
5. **Visualization** → ≥ 8 plots across Matplotlib, Seaborn, Plotly.
6. **Insights Generation** → Content, user, behavioral, and sentiment analysis.

Key Insights

Content Performance

- **Photos** → highest likes & shares.
- **Videos** → strongest watch time.
- **Lifestyle/Travel** → best performing category.
- **India & USA** → highest engagement rates; Brazil & Indonesia emerging.

User Trends

- **Age 18–24** → most active, driving nearly half of engagement.
- **Verified accounts** → higher impressions, but similar engagement rate to non-verified.

Behavioral Insights

- **Best posting time:** 6–9 PM (secondary peak at 12–2 PM).
- **Device impact:** Mobile dominates engagement; desktop users watch longer.

Sentiment Analysis

- **Positive sentiment** → highest engagement overall.
- **Negative sentiment** → more comments (debate/discussion).
- **Neutral sentiment** → moderate performance.

Conclusion

The analysis confirms that **content type, demographics, device usage, posting time, and sentiment** are critical drivers of engagement. Organizations should prioritize **visual content (photos/videos), lifestyle/travel themes, evening posting schedules, mobile optimization, and positive sentiment messaging** to maximize reach and interaction.